Move With Me: Designing Wearable Technology to Encourage Physical Activity in Rural South African Children

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***Abstract*—Rural South African children often face systemic barriers to physical activity, including unsafe outdoor environments, under-resourced Early Childhood Development (ECD) centers, and increasing exposure to sedentary screen time. This paper introduces a wearable superhero-themed Tangible User Interface (TUI) co-designed with rural caregivers to support playful movement in young children. The cape integrates motion sensors, LED feedback, Bluetooth Low Energy (BLE), and a solar-rechargeable battery, ensuring functionality in low-resource settings without the need for internet or grid electricity. A companion Android-based mobile app enhances engagement through culturally tailored animations that respond to detected movement. Grounded in co-design and informed by technology probes, this work emphasizes inclusive design practices and demonstrates how low-cost, embedded systems can support early childhood development in under-served regions. Our findings contribute to the field of emerging educational technologies, highlighting the potential of context-aware, offline-first solutions for equitable innovation.**

***Index Terms*—HCI, Co-Design, Tangible User Interface, HCI4D, Community-Based**

I. Introduction

In many parts of rural South Africa, young children face significant barriers to healthy physical movement due to a lack of safe recreational spaces and limited access to centers for early childhood development (ECD) with adequate resources (1). This challenge is compounded by the increasing prevalence of screen time, which often replaces active play and contributes to early sedentary behaviors during a formative stage of physical, emotional, and cognitive growth (1; 2; 3).

Although technology is often criticized for fueling inactivity, it can also be repurposed as a powerful tool for re-engaging children in physical play. Commercial products like the Nintendo Wii and Xbox Kinect have successfully shown how digital feedback can promote movement. However, these systems are highly dependent on high-end infrastructure and infrastructure: stable power supply, Internet connectivity, and affordability, making them inaccessible in many rural settings (4).

To address this gap, we introduce an alternative, a wearable superhero-themed cape designed as a tangible interface to motivate movement and exercise among young children. Unlike traditional technology-first solutions, this project adopts a community-driven, co-design approach. Local mothers participated in iterative workshops using a technology probe to shape the form and function of the wearable based on the real needs and behaviors of children in their communities.

The resulting cape integrates off-line-capable hardware components such as motion sensors, Bluetooth Low Energy (BLE), and solar-rechargeable power, ensuring that the system remains functional in low-resource environments. By combining playful and culturally relevant aesthetics with embedded technology, our wearable superhero cape offers an accessible and meaningful way to encourage physical activity without the need for the Internet or expensive hardware.

The remainder of this paper reviews the relevant literature, outlines our methodological approach to co-design and probe deployment, and presents the technical development of the wearable system. We conclude with reflections on user feedback and the broader implications of designing for underserved contexts.

II. Background and Related Work

A. The Growing Challenge of Sedentary Behaviour in Early Childhood

Childhood inactivity is a growing public health problem in South Africa, where many children, especially in low-income and rural settings, are leading increasingly sedentary lifestyles. Factors such as urban sprawl, safety concerns, and widespread use of screens have replaced outdoor play with television viewing, mobile gaming, and other passive behaviors (5). This shift is particularly worrying during early childhood, a critical period for establishing healthy habits and supporting physical, emotional, and cognitive development.

Recognizing the importance of this developmental stage, South Africa's national Early Childhood Development (ECD)

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policy emphasizes a holistic model that supports children from birth to nine years old. This includes physical well-being as a foundational component of long-term learning and success (6). Despite this policy, systemic challenges, such as overcrowded classrooms and limited resources, make it difficult to consistently incorporate physical activity into daily learning experiences in rural centers for ECD.

B. Tangible and Wearable Interfaces for Promoting Active Play

In recent years, Tangible User Interfaces (TUIs) have emerged as a promising design approach for encouraging physical interaction in educational and developmental contexts. TUIs enable children to learn by doing, engaging their bodies as well as their minds. Furthermore, wearable TUIs—digital systems embedded into clothing or accessories—are especially appealing for younger users, as they seamlessly blend technology with play.

One example is MakerWear, which introduced a modular electronic garment system where children can snap on sensors and lights to create interactive clothing (7). This hands-on approach inspires creativity and bodily movement while introducing basic computing concepts. Similarly, TangToys are soft, connected devices designed to support emotional expression and social interaction through tangible play, helping children communicate through gesture-based, Bluetooth-enabled responses (8).

In addition to MakerWear and TangToys, several other interactive toys have been developed to promote physical activity and creative learning through tangible interaction. For example, the KIBO Robot Kit allows children to build and program robots using wooden coding blocks, supporting movement-based, screen-free learning. Leka, a rolling robotic toy, responds to touch and motion to help children with developmental delays engage in social and physical play. Sound Blocks and Playful Bits use physical pieces that produce sound or visual effects when arranged or moved, encouraging collaboration and hands-on exploration. These examples highlight how tangible and wearable technologies can support movement, creativity, and emotional expression—offering playful, interactive alternatives to screen-based learning (9)

It is seen that TUIs can successfully merge developmental goals with interactive, engaging technology particularly when designed for play-based environments like homes or preschools. In educational settings, such tools have been shown to increase engagement and encourage physical activity among children, offering a potential solution to the growing concern of sedentary lifestyles.

C. Designing Within South Africa's Rural Technological Landscape

Despite the promise of TUIs, many commercially available products remain inaccessible to rural families in South Africa due to both economic and infrastructural barriers. With median rural household incomes around R95,770 annually (roughly USD 5,000) (10), the cost of fitness trackers, smart toys, and gaming consoles is often prohibitive. These systems also rely heavily on stable internet and electricity, both of which are unreliable and costly in rural communities (11; 12).

Beyond technical limitations, these tools are rarely designed with local cultural contexts in mind. Imported technologies often fail to align with the lived experiences of rural children and caregivers, resulting in poor adoption or disinterest (13; 14; 15). This underscores the need for low-cost, offline-capable, and locally relevant designs that consider the social and infrastructural realities of South African communities.

One proposed solution is the use of Bluetooth Low Energy (BLE) technology, which enables device-to-device communication without requiring data or Wi-Fi connectivity. BLE allows for interaction between wearables and mobile applications while keeping power consumption and operational costs low—features critical in underserved environments (16).

In this paper, we build on existing local research and present a wearable cape co-designed with mothers from a rural South African community. This solution was developed to directly address technological, financial, and cultural barriers and to explore how locally informed design processes can lead to inclusive, sustainable technology that encourages active play. The following sections detail our co-design process, the development of the prototype, and reflections on community-driven innovation in resource-constrained environments.

III. Methodology

A. Study Context and Community Setting

This research was conducted in SW* [omitted for anonymity], a rural region in KwaZulu-Natal, South Africa, located approximately 100 km from Durban. The area is predominantly Zulu-speaking and is characterized by limited infrastructure, with many households earning less than 2,500 ZAR (approximately $130 USD) per month. Sweet Waters has an established relationship with a public research institution that facilitates safe and ethical community engagement. The institution also supports a cooperative of quilters who contributed to this study by producing the capes and mentoring participating mothers in basic sewing techniques.

Ten women participated in this study, including eight mothers and two quilters. These women had previously been involved in related participatory technology design projects and were selected due to their familiarity with the concepts and tools introduced. Workshops were facilitated with the support of trusted community liaisons and conducted in a shared community space. Table I summarizes the sessions and attendance.

B. Participatory Design Approach

Our methodology was grounded in a participatory design framework comprising of co-design readiness, co-design, and co-creation. This is a layered approach chosen not only to guide the research but to empower participants as active collaborators rather than passive recipients. This was especially

Date	Workshop Focus	Attendance
31-Oct-24	Introduced functional cape probe; initial feedback	8 (M=5, Q=2, R=1)
14-Nov-24	Sewing capes, embedding initial LEDs	10 (M=6, Q=2, R=2)
28-Nov-24	Assembly of battery components	9 (M=6, Q=1, R=2)
03-Dec-24	Power calibration and testing	12 (M=8, Q=2, R=2)
12-Dec-24	Microcontroller programming, accelerometer integration	11 (M=8, Q=1, R=2)
19-Dec-24	Full electronic assembly including Neopixel LEDs	12 (M=8, Q=2, R=2)
16-Jan-25	Final testing and personalization of capes	9 (M=6, Q=2, R=1)
Key: M = Mothers, Q = Quilters, R = Researchers		

TABLE I
SUMMARY OF WORKSHOPS, OBJECTIVES, AND ATTENDANCE

important given the technological unfamiliarity, historical marginalization, and infrastructural challenges present in rural South African contexts (14). Each phase of the framework was purposefully selected to build confidence, ensure inclusivity, and foster shared ownership of the final product.

Co-design readiness was essential to bridge the digital divide and level the playing field between researchers and participants. Many rural participants had limited prior exposure to electronics or programming. By implementing a structured six-month training program using Arduino sensor kits and hands-on experimentation with circuits and code, we ensured participants had time to gain not only basic technical proficiency but also the confidence to engage critically with design decisions. This phase helped dismantle power hierarchies common in design research and positioned participants as informed contributors.

Co-design, the second phase, emphasized partnership and mutual learning. We prioritized cultural responsiveness and adaptability, allowing our methods to evolve based on the participants' lived experiences. Instead of enforcing rigid agendas, we encouraged iterative feedback and spontaneous ideation. This ensured that the wearable cape was not merely functional, but also resonated with the social norms, aesthetic preferences, and parenting values of the mothers involved. By treating them as equal co-creators, we acknowledged their contextual knowledge as vital to the artifact's success.

Co-creation marked the final phase, translating ideation into tangible outcomes. Here, participants moved beyond discussion to hands-on construction—cutting fabric, sewing circuits, connecting LEDs, and assembling the wearable system themselves. This physical engagement reinforced their sense of agency and pride in the technology, anchoring the artifact not only in technical design, but in emotional and cultural significance. Moreover, it enhanced sustainability: by understanding how the system was built, participants are more likely to maintain, adapt, or replicate the technology in their community long after the study concludes.

Ultimately, this three-part participatory approach was not only a method but a value system. It prioritized capacity building, trust and equity, ensuring that the final prototype was not imposed on the community, but rather co-developed, co-owned, and contextually grounded in their everyday realities.

C. Technology Probes as Design Catalysts

To facilitate deeper engagement and lower the cognitive load often associated with open-ended discussions, we implemented a technology probe. This was a rough, functional prototype of the superhero cape. This early version incorporated core components such as a motion sensor, Bluetooth-enabled micro-controller, and LED lights. Technology probes serve as exploratory artifacts designed not to test usability, but rather to provoke thought, gain feedback and support imaginative speculation around future functionality and use in the real-world settings.

In our context of the study, the probe functioned as a conversation starter, bridging the abstract nature of technical systems with the everyday lived experiences of these mothers in a rural setting. Rather than asking participants to ideate from scratch, which can be overwhelming or intimidating especially when dealing with unfamiliar technologies, the probe gave them something concrete to touch, test and critique. It allowed participants to visualize how the cape could fit into their children's routines, provided insight into what did or didn't resonate with them and made gaps in functionality or cultural relevance easier to articulate.

More importantly, it shifted the dynamic of the design process, by empowering the mothers to move from being passive recipients of technology to active co-designers. Their interaction with the prototype inspired discussions about movement patterns, feedback mechanisms, safety considerations, and aesthetic preferences. It also enabled participants to propose better modifications, ensuring that future iterations of the cape would better reflect their children's needs, local environmental conditions and socio-cultural values.

Ultimately, the technology probe was not just a design artifact. It was a participatory catalyst. It fostered creative exploration, made invisible ideas visible and played a critical role in translating abstract technological concepts into tangible, culturally relevant possibilities.

D. Mobile Application Feedback Session

Conducting a dedicated feedback session on the accompanying mobile application, was essential to ensure the technology ecosystem supporting the cape was both engaging and culturally appropriate. The app serves as a key interaction point for children, translating physical movement detected by the cape into meaningful digital experiences. By involving mothers directly in reviewing and shaping the app's design elements such as the character visuals, color schemes, voice prompts, and suggested movements, we ensured that the interface resonated with the local context and user preferences.

Providing blank templates for sketching allowed participants to actively express their ideas in a visual and tactile manner,

reducing barriers posed by language or technical literacy. This hands-on approach fostered deeper involvement, encouraging mothers to personalize the app in ways that would motivate their children effectively. Incorporating their feedback into UI iterations also demonstrated respect for their insights and reinforced a co-creative design idea.

Ultimately, this process helped bridge the gap between physical and digital components of the study, improving usability, cultural relevance and user satisfaction. It ensured the app was not an afterthought but an integrated part of a holistic, user-centered system designed to encourage better physical activity among rural South African children.

E. Thematic Analysis of Co-Design Sessions

All sessions were audio recorded with permission and later transcribed, anonymised, and securely stored in compliance with institutional data protocols. We employed reflexive thematic analysis (17) to examine transcripts, participant sketches, and observational notes. Our goal was to surface recurring patterns in how participants envisioned the cape's function, interaction, and visual appeal. This approach helped us refine both the digital and physical components based on real-world insights and contextual priorities.

F. Ethical Considerations

Ethical clearance was granted under protocol HREC 378/2021. All participants provided informed consent, with additional verbal confirmation obtained during media recording. Participation was voluntary, and individuals were free to withdraw at any time. Participants were involved in decision-making regarding how their likenesses and contributions were used. Many expressed pride in their participation and requested that their faces remain visible in documentation. As recognition of their contribution, mothers received honorariums and were gifted the completed capes for their children as a gesture of appreciation..

G. Study Limitations

This study offers promising insights into community-driven wearable design but also encountered several challenges.

Despite participants undergoing a readiness phase that included hands-on training, some mothers remained hesitant when working with electronic components, particularly with the batteries and wiring. A mother expressed herself saying:

> " I don't like the batteries, I am scared of them"

This hesitation may have constrained their full engagement and potential in the more technical aspects of the co-creation process. Additionally, language barriers present occasionally posed challenges. Although efforts were made to simplify technical terminology and provide interpretation support, the absence of a dedicated, full-time translator sometimes hindered clear communication of complex concepts. The design and outcomes of the project are also highly context-specific, being tailored to the Early Childhood Development (ECD) landscape of rural South Africa. Therefore, broader applicability may require significant adaptation to suit other cultural, geographic or age-specific contexts. The wearable cape was intentionally not designed for infants under 12 months due to safety concerns, limited motor development, and sizing constraints which restricts its use to children with more developed motor skills. This study involved a small, purposively selected group of ten participants, all of whom were mothers or quilters from a single rural region. While this ensured deep, context-rich engagement, it limits the generalisability of our findings. Future research could test the cape with a broader, more diverse group including fathers, teachers and children from other rural regions or age groups to assess scalability and broader usability.

IV. Findings

A. Design decsions from the Co-Design Process

1) Superhero Play to Encourage Movement: The mothers involved in the study continuously repeated how the idea of a superhero cape was effective in broadening children's imaginations. The use of the cape played a part in encouraging physical activity through role play, with the superhero identity bridging a narrative between this fantasy and their physical movement. This effectively transformed the idea of exercise into play. Playful engagement such as those observed was made possible through the cape which provided a low-cost, imaginative way to support movement outdoors, as well as indoors. One mother mentioned the following:

> "My baby, shes a girl but she adores superman and batman, so by playing with the cape she will be enjoying being a superhero and learn at the same time, cause it must be educational as well"

2) Visual Stimulation in Encouraging Movement: As included in the embedded features, children and mothers consistently responded most positively to the LED lights. The appearance of real-time visual feedback was described as exciting, responsive and easy to interpret. This is particularly more important for children who are not yet fully verbal, and therefore respond better to visual stimuli as produced by the lights. One mother offered the following:

> "My kids are drawn to lights, I think adding the lights will make it exciting for them"

3) Durability and Washability: There was concerns regarding the cape's usability on a daily basis, which included how often it can be washed and whether the electronics would be damaged. Therefore, the solar charging and detachable electronics pouch was essential in maintaining hygiene and usability, while also promoting ease of use in contexts where electricity is not always guaranteed.

B. The Relationship between Workshop Insights to the Design Outcomes

TABLE II
CO-DESIGN THEMES AND CORRESPONDING DESIGN RESPONSES

Theme	Participant Feedback	Design Outcome
Superhero Play	"She adores superman and batman, so by playing with the cape she will be enjoying being a superhero."	Used cape fabric and superhero visuals to encourage imaginative movement in familiar designs
Lights vs. Sounds	"My kids are drawn to lights, I think adding the lights will make it exciting for them"	Prioritised responsive LED strip which was more flexible; ensured audio is received when a movement is triggered
Durability & Washability	The cape must be washable due to kids playing in them as well as to promote hygiene	Modular design: detachable electronics in ventilated pouch
Electricity Access	"We don't always have electricity."	Solar-rechargeable battery and offline BLE capability
Simplicity	"I don't like the batteries, I am scared of them"	Simplified creation process where better exposure with batteries and electronics was shown

C. Co-Designed Wearable Cape: From Concept to Real-World Use

The cape was co-created through iterative collaboration with rural mothers to balance technical capabilities with the mothers' comfort and to ensure cultural relevance. Central to its design is an ADXL345 accelerometer, a compact and affordable sensor capable of accurately capturing three-axis motion data, which is crucial for recognizing different child movements, such as running and jumping, to subtle gestures. In addition, this sensor's ability to detect orientation helps the system understand how the cape is worn, a feature critical for ensuring feedback only triggers during intentional play rather than accidental bumps.

To enhance reliability in a dynamic environment, the system incorporates adaptive thresholds calibrated to typical child activity patterns collected during lab tests. The system is calibrated to respond effectively to a range of typical child energy levels and movement styles, which helps make the technology more inclusive and user-friendly.

D. Development of the Wearable Cape

1) Motion Sensing and Detection: Building upon previous iterations, our prototype integrates the ADXL345 accelerometer sensor[1] to capture three-dimensional movement data. This sensor tracks acceleration along the X, Y, and Z axes, enabling detection of a range of child movements including jumps, lateral shifts, and changes in posture. Importantly, it also senses the cape's orientation—whether it is worn upright or tilted—providing contextual awareness of how the garment is used.

To differentiate meaningful movement from accidental noise or small bumps, the system employs a combination of baseline thresholds calibrated from typical child activity, adjustable sensitivity settings tailored to individual movement patterns, and motion classification algorithms that analyze acceleration spikes (e.g., sharp increases along the Z-axis to detect jumps). This multi-layered approach ensures accurate recognition of diverse movements and enables timely, context-appropriate feedback to engage the child.

To differentiate meaningful physical activity from incidental or minor vibrations, we implemented a multi-layered detection strategy:

- **Baseline Calibration:** We established threshold values derived from common child activities such as walking, running, and jumping, setting a minimum activation level to filter out noise and accidental bumps.
- **Customizable Sensitivity:** Recognizing individual variability in children's movement patterns and energy, the system supports fine-tuning of sensitivity parameters for personalized responsiveness.
- **Activity Recognition Algorithms:** These process raw accelerometer data to classify distinct movements. For example, a jump is characterized by a sharp spike in vertical acceleration (Z-axis), which triggers specific system responses. The algorithms are designed to accommodate natural movement differences, ensuring accurate detection and feedback.

2) Central Processing Unit: At the heart of the system lies the ESP32 microcontroller, chosen for its integrated Bluetooth Low Energy connectivity and energy-efficient design which is vital for use in rural settings with intermittent power supply. The core processing is handled by this ESP32 microcontroller[2]. The microcontroller continuously collects sensor data, performs noise filtering, applies the movement classification algorithms, and manages output controls including LED lighting and audio feedback. BLE connectivity allows seamless, real-time communication with the companion mobile application, facilitating an interactive user experience without requiring Wi-Fi or cellular networks.

3) Interactive Feedback to Promote Physical Activity: To encourage ongoing child engagement and motivate movement, the wearable cape delivers multi-sensory feedback:

- **Auditory Stimulation:** An embedded MP3 module with a small speaker plays a variety of pre-recorded sounds and voice prompts stored on a microSD card[3]. Positive reinforcements like cheering and playful cues are triggered immediately upon detecting targeted movements.
- **Visual Stimulation:** A strip of flexible WS2812B Neopixel LEDs[4] embedded along the cape illuminates

[1]https://dc3dprinters.co.za/product/adxl345-3-axis-tilt-digital-gravity-acceleration-sensor/

[2]https://www.digikey.co.za/en/products/detail/espressif-systems/ESP32-DEVKITC-VIE/12091811

[3]https://www.dfrobot.com/product-1121.html

[4]https://www.diyelectronics.co.za/store/led-strips/1846-addressable-rgb-led-strip-30m-ws2812b-5v-dc-ip65-casing.html

with vibrant, dynamic color patterns in response to movement, providing an engaging and tangible reward for activity.

4) Material and Construction Considerations: The cape is crafted from a soft, lightweight, and breathable fabric to ensure comfort during active play, while maintaining durability for everyday wear. To engage children, the fabric features vibrant superhero-themed graphics that resonate culturally and visually with our target age group. Crucially, the electronic modules including sensors, LEDs, and battery packs are housed in a detachable, ventilated compartment. This modular approach allows caregivers to easily remove the electronics, enabling the cape itself to be machine-washed without risk of damage. Such design considerations are essential to maintaining hygiene and extending the lifespan of our wearable, particularly in rural environments where resources for replacement of these components may be limited.



Fig. 1. Prototype of the superhero cape used as a technology probe during co-design workshops

E. Mobile Application: Engaging Children Through Culturally Relevant Gamification

The companion mobile application, developed using Kotlin for Android's widespread adoption in South Africa, transforms physical movements into a fun, interactive game experience. Through Bluetooth Low Energy communication, the app receives live motion data from the cape, allowing it to provide instant animated responses.

Designed with input from the mothers, the app features characters and narratives that reflect local cultural elements, enhancing children's connection and motivation. Movement prompts encourage specific activities, and successful completion triggers rewarding animations and sounds, reinforcing positive behavior.

The app functions fully offline, a critical consideration given the unreliable internet access in rural areas. It stores session data locally and syncs with a server only when connectivity is available, balancing data integrity with accessibility and privacy.

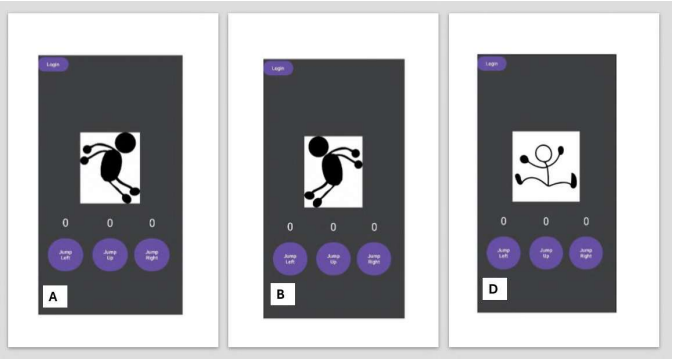


Fig. 2. Screenshot of the app showing animated feedback upon detection of a jump

F. Feedback from Participant Interaction and Real-World Use

Feedback from mothers and children during co-design workshops highlighted the importance of intuitive responsiveness and physical comfort. Participants appreciated the cape's bright, playful lights and immediate audio cues, noting increased enthusiasm in children's movement. However, some raised concerns about battery life and the ease of charging in areas with limited electricity.

The iterative design process allowed continuous refinement based on this input, leading to improved circuit robustness, optimized power consumption, and user-friendly interfaces. These participatory insights were vital for creating a wearable system that not only functions well technically but also fits seamlessly into the daily lives of rural South African families.

V. Discussion

Our research demonstrates the potential of co-designed wearable technology to address early childhood inactivity in under-resourced environments by considering culturally and contextually relevant features with tangible interfaces and locally adaptable technology. Our wearable cape system thus offers a potential, scalable and contextually appropriate response to the sedentary behavior crisis seen in rural South African communities.

A. From Global Innovation to Local Relevance

The findings validate many of the benefits previously attributed to Tangible User Interfaces (TUIs) and wearable technologies, such as those exemplified by MakerWear, TangToys, and KIBO (7; 8; 9). These global projects highlight how physical interaction promotes creativity, movement and engagement among young users. However, their limited availability and high cost make them inaccessible in many Global South contexts. Our work builds on this body of knowledge by reimagining similar interactive principles within a design ecosystem that is constrained by affordability, electricity reliability and offline usability, which are often conditions typically seen in rural South Africa.

B. Embedding Co-Design in Technological Adaptation

Central to the success of our solution was the co-creation process with mothers, which not only shaped the visual and physical characteristics of the cape but also addressed practical elements such as its charging solutions, the modular design for washability and app content. While many commercial TUIs are developed with Western educational settings in mind, our work proves the necessity of participatory design in ensuring technological solutions are usable, accepted and sustainable in diverse contexts (13; 14).

The feedback loop between users and designers was crucial in shaping a system that feels intuitive and meaningful to children and caregivers. Rather than introducing a foreign or intimidating technology, the cape was seen as a playful extension of everyday life, aligning with local cultural values and their parenting routines.

C. Supporting Physical Activity Through Multi-Sensory Feedback

Our integration of the ADXL345 accelerometer, combined with custom motion classification algorithms and adjustable sensitivity, demonstrated how low-cost sensors can provide movement recognition suitable for stimulating physical activity. The multi-sensory feedback using LED lighting and triggering audio, proved to be an effective motivational tool, echoing findings from prior work in physical interaction and gamified learning.

Importantly, our findings show that contextualised feedback and not just generic responses help us reinforce behavior. For instance, audio cues featuring local languages or familiar sounds can boost engagement. This supports the idea that cultural familiarity contributes to the effectiveness of TUIs in educational and developmental settings within the rural areas.

D. Infrastructure Constraints Reduced With BLE and Offline Capability

Our major system innovation is in the design of being able to operate the system completely offline due to the use of Bluetooth Low Energy (BLE), which addresses the infrastructural constraints faced by these rural communities. Our solution operates in contrast to many applications and consoles that do require constant connectivity. The mobile app, which is built using Kotlin, allows for data sync updates when connectivity is reached, thus providing greater accessibility.

Our design opens the way for adaptation in similar rural regions, not only within South Africa, but also across the world. By using energy-efficient components like the ESP32, it highlights the possibility for being able to create interactive systems like ours that are both robust and sustainable, which is necessary for use in these under-served communities.

E. Designing For Inclusivity

The results from our study did highlight the use of embedding this technology in a way that feels natural and comfortable, instead of unfamiliar for both the mothers and the children. By focusing on their comfort, easy maintenance and child-friendly visual appeal, we aimed to make the wearable cape both practical and meaningful. This supports the idea that educational technology should not only be accessible, but also help children feel that they belong (15).

The superhero-themed design of our cape, along with the animations in the app that reflect the local culture, made the experience fun and familiar for them. This shows how technology in early childhood can also support creativity and movement rather than just delivering instructions or content.

Overall, our work demonstrates that through thoughtful integration of emerging technology, co-design methodologies and desining with infrastructural awareness in mind, it is possible to deliver interactive, inclusive learning systems that respond to real-world challenges in early childhood development. Our wearable cape represents a combination of technical innovation and cultural relevance which sets a foundation for future research in inclusive and community-based wearable technology.

VI. Conclusion

This study presents the design and implementation of a wearable tangible interface specifically crafted to encourage active play among young children in rural South African communities. Our findings emphasize that addressing physical inactivity in such settings demands solutions grounded in local realities both cultural and infrastructural. By actively involving mothers and caregivers through co-design and co-creation, we encouraged a sense of ownership and ensured that the technology resonates with the users' everyday experiences and needs.

The integration of low-power wireless technologies like Bluetooth Low Energy (BLE) was critical to overcoming connectivity challenges typical of rural environments, enabling real-time interaction without reliance on constant internet access. The modular design of the wearable, combined with a culturally sensitive mobile application, provides a sustainable and adaptable platform that can support early childhood development through engaging and movement-based play.

Ultimately, this work demonstrates the value of community-centered, context-aware approaches in developing technological interventions for under-served populations. Such methods not only produce more effective and meaningful tools but also contribute to empowering local communities in shaping their own health and well-being futures. Moving forward, this foundation can guide further innovations aimed at bridging technology gaps and promoting healthier lifestyles for children in similar low-resourced contexts worldwide.

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